HIERO-CODE · SYMBOLIC COMPUTE



The Reference Codex

A field guide to the rune & numeral systems — Egyptian, Sumerian, Mayan, Khoisan, and the Pyri emoji layer.



Hiero-Code treats ancient scripts as computational alphabets. Each system below is a different way of packing meaning into a few repeating marks — base-10 sums, base-60 places, base-20 stacks, parallel click channels — and the Pyri system extends that same instinct from counting quantity to encoding emotion.

I·EGYPTIAN II·SUMERIAN III·MAYAN IV·KHOISAN
V·GLYPH LIBRARY VI·PYRI EMOJI VII·EMOJI LINEAGE

Egyptian Hieroglyphic

Base 10 · Additive

Ancient Egyptian decimal notation using pictographic symbols. Each power of ten has a distinct hieroglyph; numbers are written by repeating and summing symbols — there is no place value. Attested from c. 3000 BC on papyrus and carved in stone.

Power Symbols

GLYPH	VALUE	NAME	MEANING
	1	Stroke	Single unit, the tally mark
	10	Heel Bone (Hobble)	The first grouping, ten strokes
	100	Coil of Rope	The coil, one hundred
	1,000	Lotus Flower	Abundance, the lotus in bloom
	10,000	Bent Finger	Power of the pharaoh's hand
	100,000	Tadpole / Frog	The infinite multiplier
	1,000,000	Kneeling God (Heh)	The god of eternity, a million

Unit Fractions — Rhind Papyrus style, "parts" written above a number

GLYPH	FRACTION	DECIMAL
	1/2	0.5
	2/3	0.6667
	3/4	0.75
	1/3	0.3333
	1/4	0.25
	1/5	0.2
	1/7	0.1429
	1/10	0.1

Worked example. 1,234 = 1×  + 2×  + 3× + 4× — read as a simple sum of grouped pictographs.

II Sumerian Sexagesimal

Base 60 · Positional

The world's first place-value system, c. 3000 BC, pressed in cuneiform wedges on clay. Base 60 is divisible by 2, 3, 4, 5, 6, 10, 12, 15, 20 and 30 — which is why it still rules our clocks (60 seconds, 60 minutes) and circles (360°).

Cuneiform Numerals

GLYPH	VALUE	NAME
	1	Aš (One)
	2	Min (Two)
	3	Eš (Three)
	4	Limmu (Four)
	5	Ia (Five)
	6	Aš (Six)
	7	Imin (Seven)
	8	Ussu (Eight)
	9	Ilimmu (Nine)
	10	U (Ten)
	20	Niš (Twenty)
	30	Ušu (Thirty)
	40	Nimin (Forty)
	50	Ninnu (Fifty)
	60	Géš (Sixty)
	600	Géš-U (Six Hundred)
	3,600	Šár (3,600)
	36,000	Šár-U (36,000)
	216,000	Šár-Gal (Great Šár)

Igi Fractions — reciprocals of the "regular" divisors of 60

RECIPROCAL	FRACTION	DECIMAL
igi-2	1/2	0.5

RECIPROCAL	FRACTION	DECIMAL
igi-3	1/3	0.3333
igi-4	1/4	0.25
igi-5	1/5	0.2
igi-6	1/6	0.1667
igi-10	1/10	0.1
igi-12	1/12	0.0833
igi-15	1/15	0.0667
igi-20	1/20	0.05
igi-30	1/30	0.0333
igi-60	1/60	0.0167

Worked example. 3,661 = 1×3600 + 1×60 + 1 → place-string `1, 1, 1` in base-60 (one  , one  , one).

III Mayan Vigesimal

Base 20 · Vertical Positional

A base-20 system stacked vertically: dots (1) and bars (5) fill each position, and a shell glyph marks zero — one of humanity's earliest written zeros. Each step up the stack multiplies by twenty.

GLYPH	VALUE	NAME	MEANING
	0	Shell (0)	The void, the placeholder — zero as a written concept
●	1	Dot (1)	Single dot — one unit
—	5	Bar (5)	Horizontal bar — five units, the hand
● ● ● ● — — —	9	Nine	Four dots over one bar — the Bolontiku underworld lords
● ● ● ● — — — —	19	Nineteen	Four dots over three bars — max single-position value
↑	20	Twenty	Positional step — triggers the next vertical layer (20^1)
↑ ↑	400	Four Hundred	Twenty squared — two layers up (20^2)
↑ ↑ ↑	8,000	Eight Thou-sand	Twenty cubed — the great calendrical cycle (20^3)

IV Khoisan Click Phonemes

Multi-Channel · Parallel

Not a number system but a parallel encoding: a single click carries 4–6 simultaneous channels of information — tongue position, airflow direction, contact placement, accompaniment and nasalization. One sound = several data streams at once.

C

Dental Click IPA |

Tip of tongue against front teeth, quickly withdrawn

Tongue position · Ingress airflow · Dental placement

Tc

Alveolar-Palatal Click IPA ‡

Tip of tongue against alveolar ridge and adjacent palate, released sharply downwards

Alveolar contact · Palatal friction · Sharp release

Q

Palatal Click IPA !

Tongue pressed against upper palate, released sharply — like a cork pulled from a bottle

Full palate contact · Pressure build · Pop release

X

Lateral Click IPA ||

Click at the side of tongue held pressed against palate

Lateral tongue · Side airflow · Palate slide

V The Glyph Library

The full inscribed set spanning numerals, deities, determinatives and phonetic signs — the raw vocabulary from which Hiero-Code rune nodes draw their pictographic identity.

Egyptian — 17 glyphs

GLYPH	NAME	VALUE	CAT-EGORY	MEANING
𓏺	Staff (1)	1	NUMERAL	Single unit, the tally mark
𓎆	Heel Bone (10)	10	NUMERAL	The first grouping, ten strokes
𓍢	Coil of Rope (100)	100	NUMERAL	The coil, one hundred
𓆼	Lotus Flower (1000)	1,000	NUMERAL	Abundance, the lotus in bloom
𓂭	Bent Finger (10000)	10,000	NUMERAL	Power of the pharaoh's hand
𓆐	Tadpole (100000)	100,000	NUMERAL	The infinite multiplier
𓁨	Kneeling God (1000000)	1,000,000	NUMERAL	The god of eternity, a million
𓋴	Nefr (Zero/Balance)	0	SPECIAL	Beautiful-perfect, balance, the base
𓂀	Eye of Horus	1	SYMBOL	Vision, protection — the Wedjat fractions 1/2…1/64
𓇳	Sun Disk (Ra)	1	DEITY	Solar identity, the source of all light
𓊽	Djed Pillar	1	SYMBOL	Stability, endurance, structural integrity
𓋹	Ankh	1	SYMBOL	Life force, the key of the Nile
𓌀	Was Sceptre	1	SYMBOL	Power and dominion
𓅞	Thoth (Ibis)	1	DEITY	Wisdom, mathematics, writing
𓀀	Man (Seated)	1	FIGURE	Unity, primal form, the first count
𓆑	Horned Viper	10	ANIMAL	Transformation, the tenth power
𓂋	Mouth (r)	1	PHONET-IC	Speech, the glyph of one, output

Sumerian — 17 glyphs

GLYPH	NAME	VALUE	CATEGORY	MEANING
𒁹	Ash (1)	1	NUMERAL	The single vertical wedge — the base unit
𒌋	U (10)	10	NUMERAL	Ten units in sexagesimal notation
𒐐	Géš (60)	60	NUMERAL	One sixty — the base unit of the cosmos
𒐖	Géš-U (600)	600	NUMERAL	Ten sixties — the administrative unit
𒐕	Šár (3600)	3,600	NUMERAL	The great cycle — 60^2, one shar
𒐊	Niš (20)	20	NUMERAL	Two tens, a score
𒐋	Ušu (30)	30	NUMERAL	Three tens, the lunar month
𒐌	Nimin (40)	40	NUMERAL	Two twenties

GLYPH	NAME	VALUE	CATEGORY	MEANING
	Ninnu (50)	50	NUMERAL	Fifty, close to the great base
	Aš (Whole)	1	COUNTING	The complete unit
	Min (Two)	2	COUNTING	Duality, paired forces
	Eš (Three)	3	COUNTING	Trinity, the triad of existence
	Ia (Five)	5	COUNTING	The hand, five fingers
	Dingir (God)	1	DETERMINATIVE	Divinity, the first determinative
	Giš (Tree/ Sixty)	60	DETERMINATIVE	The great wood, also sixty — structural support
	Enki (Wisdom)	40	DEITY	God of wisdom, water, magic — 40 in the pantheon
	Enlil (Air)	50	DEITY	God of wind and storm — 50 in the pantheon

Mayan — 8 glyphs

GLYPH	NAME	VALUE	CATEGORY	MEANING
•	Mayan Dot (1)	1	NUMERAL	Single dot — one unit in base-20
—	Mayan Bar (5)	5	NUMERAL	Horizontal bar — five units, the hand
	Mayan Shell (0)	0	NUMERAL	The shell glyph — zero, the void, the placeholder
• • • • — — —	Mayan 9	9	NUMERAL	Four dots + bar — the Bolontiku underworld lords
• • • • — — — —	Mayan 19	19	NUMERAL	Four dots + three bars — max single-position value
↑	Mayan 20 (Next Layer)	20	NUMERAL	Positional step — triggers the next vertical layer
↑ ↑	Mayan 400	400	NUMERAL	Twenty squared — two positional layers up
↑ ↑ ↑	Mayan 8000	8,000	NUMERAL	Twenty cubed — the great calendrical cycle

VI The Pyri System

The EMOJI Layer · Valence × Arousal × Gesture

Where the older scripts count quantity, Pyri encodes affect. It is the keyboard-native emoji of Hiero-Code: QUERTY-typeable emoticons that serve as macro-nodes for emotional state in machine-learning embeddings.

Why an emoji system belongs beside the numerals

Every other script in this codex answers the question “how many?” The Pyri system answers a different one — “how does the signal feel?” Yet it is built from the exact same architectural instinct that makes the numeral systems powerful: **a tiny alphabet of primitives, combined by rule, spanning an enormous expressive space.** Egyptian repeats seven pictographs; Sumerian positions a handful of wedges across powers of sixty; Pyri takes **12 eye-forms × 8 mouth-forms × 8 gestures** and multiplies them into **768 distinct macro-expressions.** The emoji is not decoration bolted onto a symbolic language — it is the same combinatorial engine pointed at emotion instead of number.

The Three Generative Axes

Eyes — 12 forms

EYE	VALENCE	AROUSAL	READS AS
`>.<`	Negative	High	Frustration / embarrassment
`^.^`	Positive	Low	Happiness / satisfaction
`0.0`	Neutral	Medium	Shock / blank stare
`@.@`	Neutral	High	Overstimulation / dizzy
`*_*`	Positive	High	Awe / love-struck
`$.$`	Positive	Medium	Greed / motivation
`-.-`	Negative	Low	Boredom / deadpan
`~.~`	Neutral	Low	Sleepy / indifference
`U.U`	Negative	Low	Defeat / resignation
`X_X`	Negative	Medium	Done / defeated
`3_3`	Neutral	Low	Dreamy / indifferent
`+.+`	Positive	Medium	Focus / interest

Mouths — 8 forms

MOUTH	SHAPE	VALENCE
.	dot	Neutral
_	flat	Neutral
3	open	Positive
O	surprised	Neutral
V	sad	Negative
W	wavy	Positive
P	tongue	Positive
~	wavy-alt	Neutral

Gestures — 8 forms

GLYPH	GESTURE
∅	None
\(-.-)/	Shrug / Surrender
\(^.^)/	Waving / Joy
(T_T)	Crying
\(O_O)/	Shock
\(*_*)/	Awe
\(>.<)/	Frustration gesture
;;	Sweat drops

The Emotion Vector — how a face becomes a number

Each composed emoji collapses to a three-component vector `[valence, arousal, gesture]`. Eyes contribute valence and arousal; the mouth nudges valence; the gesture adds a motor / embodiment code. Valence and arousal together are the classic circumplex model of affect — the same two axes psychologists use to map every human feeling onto a plane. Pyri adds a third, bodily axis so a node can express not just what it feels but what it is doing about it.

Canonical Emotion Vectors

`*_*`

High Arousal Positive

V +1 | A +1 | G 0

Awe / love-struck — strong engagement modeling

`^.^`

Low Arousal Positive

V +1 | A -1 | G 0

Contentment / pleasure — reward signal

`>.<`

High Arousal Negative

V -1 | A +1 | G 0

Frustration — mild distress node

`U.U`

Low Arousal Negative

V -1 | A -1 | G 0

Defeat / resignation — passive state

`@.@`

Neutral Overstimulation

V 0 | A +1 | G 2

Sensory overload — trigger scanning node

`3_3`

Neutral Dormant

V 0 | A -1 | G 0

Fatigue / passive — low-energy node

`$.$`

Motivated

V +1 | A 0 | G 1

Incentive / motivation — goal-seeking node

`\(-.-)/`

Surrender Gesture

V -1 | A 0 | G 3

Motor-expression mapping — gestural node

Seeded Pyri Glyph Nodes

GLYPH	NODE	V	A	G	MEANING
`>.<`	Frustration Node	-1	+1	0	High arousal, negative valence. Frustration / embarrassment.
`^.^`	Contentment Node	+1	-1	0	Low arousal, positive valence. Happiness / satisfaction.
`0.0`	Shock Node	0	+1	0	Medium-high arousal, neutral valence. Attention / focus trigger.
`@.@`	Overload Node	0	+2	2	High arousal, neutral-negative. Sensory overload detection.
`*_*`	Awe Node	+1	+2	0	High arousal, positive. Strong engagement / awe.
`$.$`	Motivation Node	+1	0	1	Medium arousal, positive. Incentive / goal-seeking drive.
`U.U`	Defeat Node	-1	-1	0	Low arousal, negative. Resignation / passive state.
`\ (-.-)/`	Surrender Node	-1	0	3	Medium arousal, negative gesture. Motor-expression mapping.

Connective Commentary — the emoji system in conversation with the others

↔ Sumerian — place value becomes axis value

Sumerian gets its reach from positional encoding: the same wedge means 1, 60, or 3,600 depending on where it sits. Pyri performs the identical trick across semantic positions instead of magnitudes. A symbol's meaning depends on which slot a primitive occupies — eye-slot, mouth-slot, or gesture-slot. Replace base-60 place value with a three-axis product space and you have the Pyri tensor: `Eyes ⊗ Mouths ⊗ Gestures`. Both systems buy a vast vocabulary with a handful of signs by letting position carry meaning.

↔ Egyptian — additive sentiment

Egyptian numerals simply add up: stack the pictographs and sum them. Pyri's valence works the same way — the eye's valence and the mouth's valence are summed and clamped to Positive / Neutral / Negative. A high-valence eye (^.^) plus a sad mouth (v) cancel toward Neutral, exactly as Egyptian symbols accumulate toward a total. Emotion, here, is arithmetic.

↔ Khoisan — parallel channels of feeling

The Khoisan click is the deepest cousin of the emoji. A single click fires 4–6 information channels simultaneously; a single Pyri glyph fires valence, arousal and gesture simultaneously. Neither is a linear stream of tokens — both pack multiple orthogonal dimensions into one indivisible mark. The emoji is, quite literally, a visual click: one glyph, many parallel readings.

↔ Mayan — a zero for emotion

The Mayan shell gave the world a written nothing — a placeholder that makes higher positions legible. Pyri keeps its own zero: the Neutral valence and the empty gesture (⌀). Neutral is not absence of data; like the shell, it is a meaningful placeholder that lets the surrounding signal be read precisely.

↔ The Rune Engine — affect as a control signal

Hiero-Code rune nodes already carry two continuous scalars: ASE vitality (influence) and Maat balance (stability). The Pyri vector slots straight into that machinery — arousal behaves like a vitality gain, valence like a balance bias, and the gesture code like a discrete action selector. The emoji layer is therefore the **human-readable face of the same numbers the engine already computes**: it makes a model's internal emotional state legible at a glance, using the one script every modern keyboard can already type.

In the Hiero-Code cosmology the ancients counted the heavens in tens, sixties and twenties; the Pyri system counts the inner weather of a mind. Same instinct, new axis — a pictographic alphabet of feeling, 768 expressions deep.

VII The Lineage of Feeling

Emoji Ancestry · 1982 – 2010

The Pyri layer did not appear from nowhere. It is the heir of three real traditions of typed feeling — Western emoticons, Japanese kaomoji, and the carrier emoji of the late-1990s and 2000s. This chapter is the historical reference behind the system.

Two keyboards, one instinct

Long before color emoji, people built faces from punctuation. In 1982 Scott Fahlman proposed `:-)` and `:-(` on a Carnegie Mellon board; Japan answered in 1986 with `(^_^)`. Two grammars emerged: the West reads sideways and weights the mouth; Japan reads face-on and weights the eyes. Pyri is deliberately the second kind — its twelve **eye-forms** carry the affect while the mouth only nudges it, exactly as kaomoji do.

Western emoticons — read sideways

EMOTICON	MEANING	PYRI ECHO
`:-)`	Smile / happy	`^.^`
`:-D`	Grin / laughing	`0.0`
`;-)`	Wink / joking	`+.+`
`:-(`	Sad / frown	`U.U`
`:-P`	Tongue out / playful	`+.+`
`:-O`	Surprised / shocked	`0.0`
`:-/`	Skeptical / unsure	`-.-`
`:'(`	Crying / weeping	`U.U`
`>:-)`	Evil grin / mischief	`>.<`
`B-)`	Cool / sunglasses	`$.$`
`:-|`	Neutral / deadpan	`~.~`
`:-*`	Kiss	`*_*`
`<3`	Heart / love	`*_*`
`</3`	Broken heart	`U.U`
`XD`	Laughing hard	`>.<`
`:-$`	Embarrassed / blushing	`@.@`
`%-)`	Dizzy / overloaded	`@.@`
`o:-)`	Angel / innocent	`^.^`

Kaomoji — read face-on (eye-weighted)

KAOMOJI	MEANING	PYRI ECHO
`(^_^)`	Happy / content	`^.^`
`(>_<)`	Frustration / wincing	`>.<`
`(T_T)`	Crying	`>.<`
`(;_;)`	Quiet sobbing	`U.U`
`(._.)`	Blank / neutral	`~.~`
`(=_=)`	Exhausted / deadpan	`-.-`
`(¬ω¬)`	Side-eye / unimpressed	`3_3`
`(♡ω♡)`	In love / adoring	`*_*`
`¯\_(•_•)_/¯`	Shrug / “I dunno”	`\ (-.-)/`
`(╯°□°)╯ ~ ┻┻`	Table flip / rage	`>.<`
`m(_ _)m`	Deep bow / apology	`U.U`
`\(^o^)/`	Celebration / joy	`\ (^.^)/`

The Carrier Emoji Era — Japan, 1997 onward

The pictographic emoji proper was born on Japanese mobile phones, where a 250-character limit forced designers to pack meaning into 12×12-pixel icons. Three carriers each shipped their own mutually-incompatible set — the original reason a single code could mean a convenience store on one network and a wristwatch on another, a problem later resolved by folding emoji into the Unicode standard.

1997	**J-Phone (later SoftBank)** 90 emoji on the SkyWalker DP-211SW — 12×12 px, monochrome. The first mobile emoji set, including an early pile-of-poo. Later shaped Apple's emoji.
1999	**NTT DoCoMo** Shigetaka Kurita's 176 emoji for the i-mode service — single-hue on a 12×12 grid. This set was later acquired by the Museum of Modern Art.
1999	**au by KDDI** Type A-2: 170 black-and-white emoji for EZweb. Gained animation in the early 2000s and later seeded Google's emoji set.
2000–03	**Carrier expansion** au Type B-2 (188 emoji), B-3 (added shading), then the first animated sets. Each carrier's emoji stayed mutually incompatible across networks.
2008	**Apple & Google** Apple Color Emoji ships on iPhone OS using SoftBank's private-use scheme; Google adds emoji to Gmail in collaboration with au by KDDI.
2010	**Unicode 6.0** Carrier emoji formally standardized into Unicode — the bridge from the fragmented carrier era to today's universal set.

Inside DoCoMo's Original 176 — Kurita's ten categories, 1999

Shigetaka Kurita's set — now in the Museum of Modern Art — organized its 176 glyphs into ten thematic families. They map cleanly onto the structure Hiero-Code uses today: most are nouns (things and places), but one whole family is pure affect — the direct ancestor of Pyri.

CATEGORY	REPRESENTATIVE CONTENTS
Weather & Nature	Sun, cloud, rain, snow, typhoon, moon, sakura, autumn leaf
Faces & Emotions	Happy, sad, angry, surprised, sleepy, sick, love-struck, cool
Hearts & Symbols	Heart, broken heart, the card suits ♠ ♥ ♦ ♣, question mark
Communication & Tech	Mobile phone, telephone ☎, pager, TV, computer, envelope ✉
Transport & Places	Car, taxi, bus, train, airplane, ship, house, hospital
Food & Drink	Noodles, bento, rice, beer, coffee ☕, cake, fruit, fast food
Sports & Activities	Soccer, baseball, tennis, skiing, music note, microphone, games
Time & Calendar	The twelve clock faces, one o'clock through twelve
Zodiac	The twelve signs ♈ ♉ ♊ ♋ ♌ ♍ ♎ ♏ ♐ ♑ ♒ ♓
Money & Business	Yen, rising chart, 100-points, OK / NEW / UP / FREE markers, recycle ♻

↔ Why this belongs in the Codex

Every ancestor above solved the same problem the numeral systems solve: a tiny set of typeable marks, combined by rule, spanning a huge space of meaning. Western emoticons and kaomoji proved a face could be built from keyboard primitives; the carrier sets proved those

> faces could be standardized and counted. Pyri inherits both lessons — it is a face-on, eye-weighted, QUERTY-typeable alphabet of emotion, descended in a straight line from `(^_^)`.

From *:-)* in 1982 to a 768-expression tensor — the same instinct, carried forward one keyboard at a time.